
Behavioral Interventions (and energy policy)

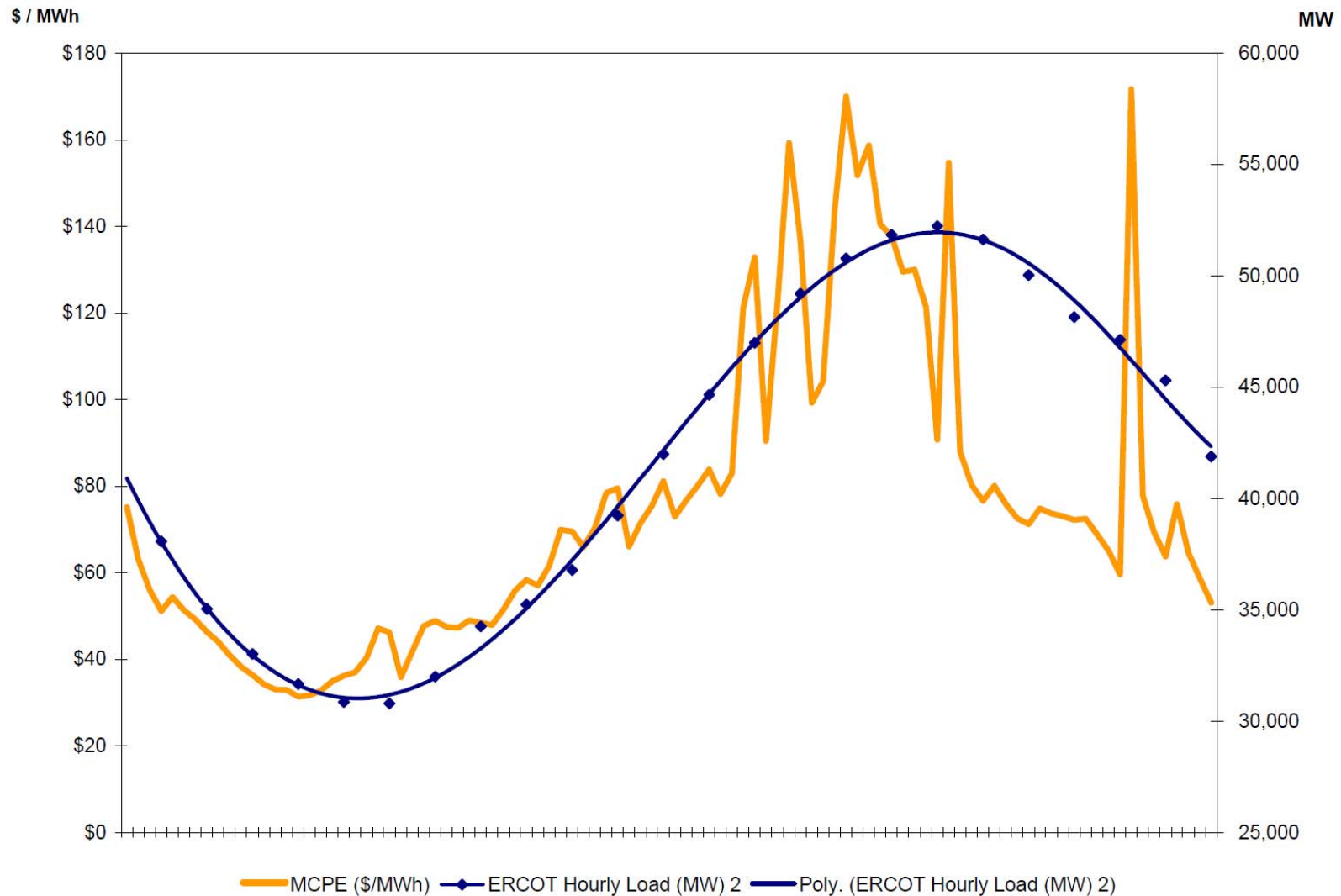
Econ 649

Standard Arsenal of Policy Instruments

1. Price

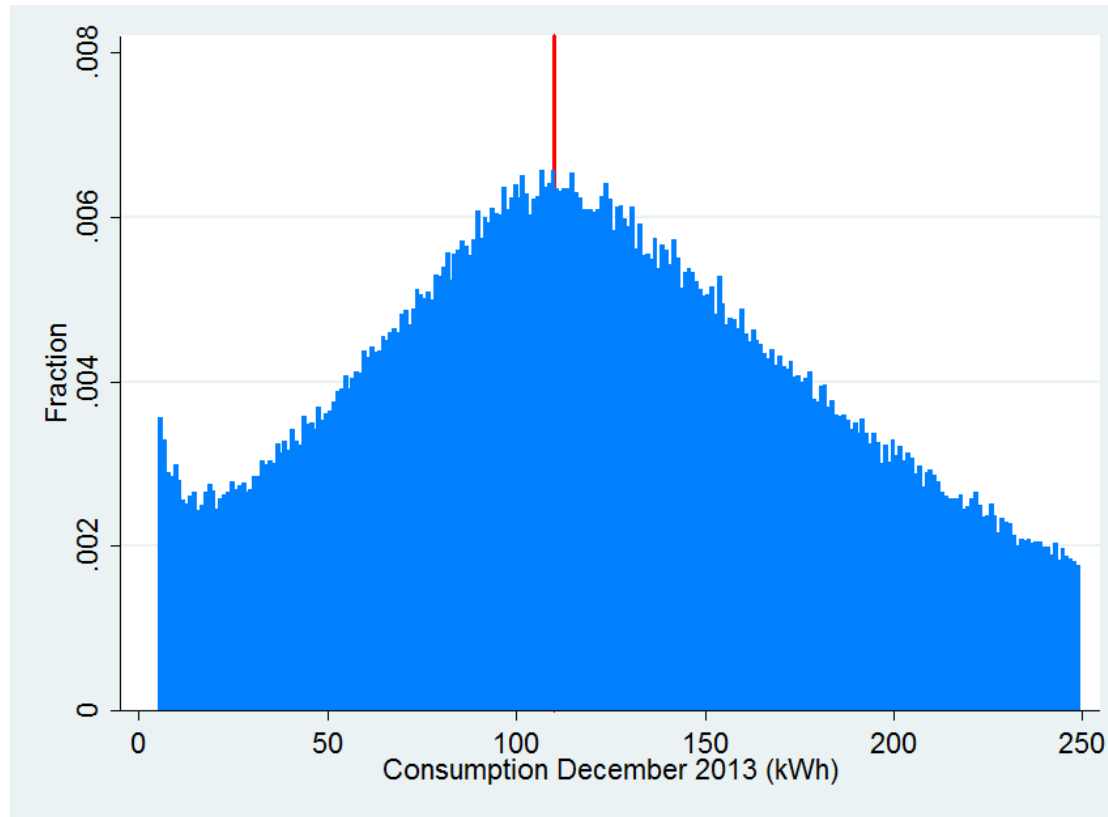
- ❑ “Get the price right”, including private and social costs
- ❑ Problems:
 1. Internalities
 2. Political costs of exposing customers to true social costs
 - ❑ Ex 1: Carbon tax
 - ❑ Ex 2: Fuel costs in developing countries
 - ❑ Ex 3: Prices in peak demand periods

Example: Wholesale Electricity Prices in Texas During the Summer



Do People Pay Attention to Electricity Prices?

- In Ecuador:
 - if you consume 110 kwh in a month, you pay about \$6.31
 - if you consume 111 kwh in a month, you pay about \$11.63





EMPRESA
ELÉCTRICA
QUITO S.A. E.E.Q.

Las Casas E1-24 y Av. 10 de Agosto R.U.C.: 1790053881001 / CONTRIBUYENTE ESPECIAL / RESOLUCIÓN Nº. 5368

Factura No. 001-007-003108291

Autorización SRI: 1110588751

Fecha de autorización: 01/02/2012

Válida hasta: 13/01/2013



No. de Control: 149805609-83

Valor a pagar: 23.24

Fecha de emisión

18/12/2012

Fecha de vencimiento

04/01/2013

INFORMACIÓN DEL CONSUMIDOR:

SUMINISTRO: 1498056 - 3 PROAÑO DURAN MARCO ANTONIO

CODIGO ÚNICO ELÉCTRICO NACIONAL: 1401498056 Cédula / R.U.C.: 170667497-3

Dirección servicio: NEW CASTLE 10 VIA GUANGOPOLLO C.H. NEW CASTLE

Dirección notificación: Domicilio

Plan/Geocódigo: 74 01-07-005-2390

Parroquia - Cantón: CONOCOTO DISTRITO METROPOLITANO QUITO

Tarifa: Residencial (Baja Tensión)

18/12/2012

SUMINISTRO DEL SERVICIO ELÉCTRICO:

Medidor: 607818-SAX-AB Factor de multiplicación: 1.00 Constante: 1.00

Recargo Pérdidas en Transformación: 0 %

Desde: 15/11/2012 Hasta: 14/12/2012 Días: 29 Tipo consumo: Leído

Descripción	LECTURAS			Valores
	Actual	Anterior	Consumo Unid.	
Energía	6362.00	6160.00	202 kWh	14.77
07h00-22h00			kWh	0
22h00-07h00			kWh	0
Reactiva			kvarh	0
Demanda Cliente			kV	0
Maxima			kV	0
Maxima en pico			kV	0

VALOR FACTURABLE: 14.77

COMERCIALIZACION: 1.41

SUBSIDIO SOLIDARIO: 1.62

SERV. ALUM. PUB.: 1.46

I.V.A. (0%): 0.00

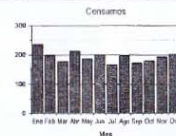
TOTAL SERVICIO ELÉCTRICO (1): 19.26

Su ahorro por la Tarifa de
la Dignidad es de
0.00

La Energía ya es de todos!

VALORES PENDIENTES DE PAGO POR SERVICIO ELÉCTRICO:

CONCEPTO	VALOR
TOTAL VALORES PENDIENTES:	



SIN EL SELLO DE CANCELADO, NO TIENE VALOR



EMPRESA
ELÉCTRICA
QUITO S.A. E.E.Q.

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Parroquia - Cantón: CONOCOTO DISTRITO METROPOLITANO QUITO

Tarifa: Residencial (Baja Tensión)

18/12/2012

OTROS VALORES A PAGAR:

CONCEPTO	SUSTENTO LEGAL	VALOR
IMPUESTO BOMBEROS	Ley de Defensa Contra Incendios	1.46
TASA RECOLECCION BAS	Ordenanza Municipal	2.52
VALORES PENDIENTES DE PAGO POR SERVICIO ELÉCTRICO:		
TOTAL OTROS VALORES A PAGAR (2):		3.98

TOTAL A PAGAR

Valor servicio eléctrico (1)	19.26
Otros valores a pagar (2)	3.98
TOTAL (1) + (2):	23.24

(*) BASE PARA RETENCIÓN 1%: 0.00

Pagar hasta: 04/01/2013

CAR UNICORNIO 001 01/04/2013 13:42:53 LECONIB_86

PROAÑO DURAN MARCO ANTONIO 1001498056017 23.24

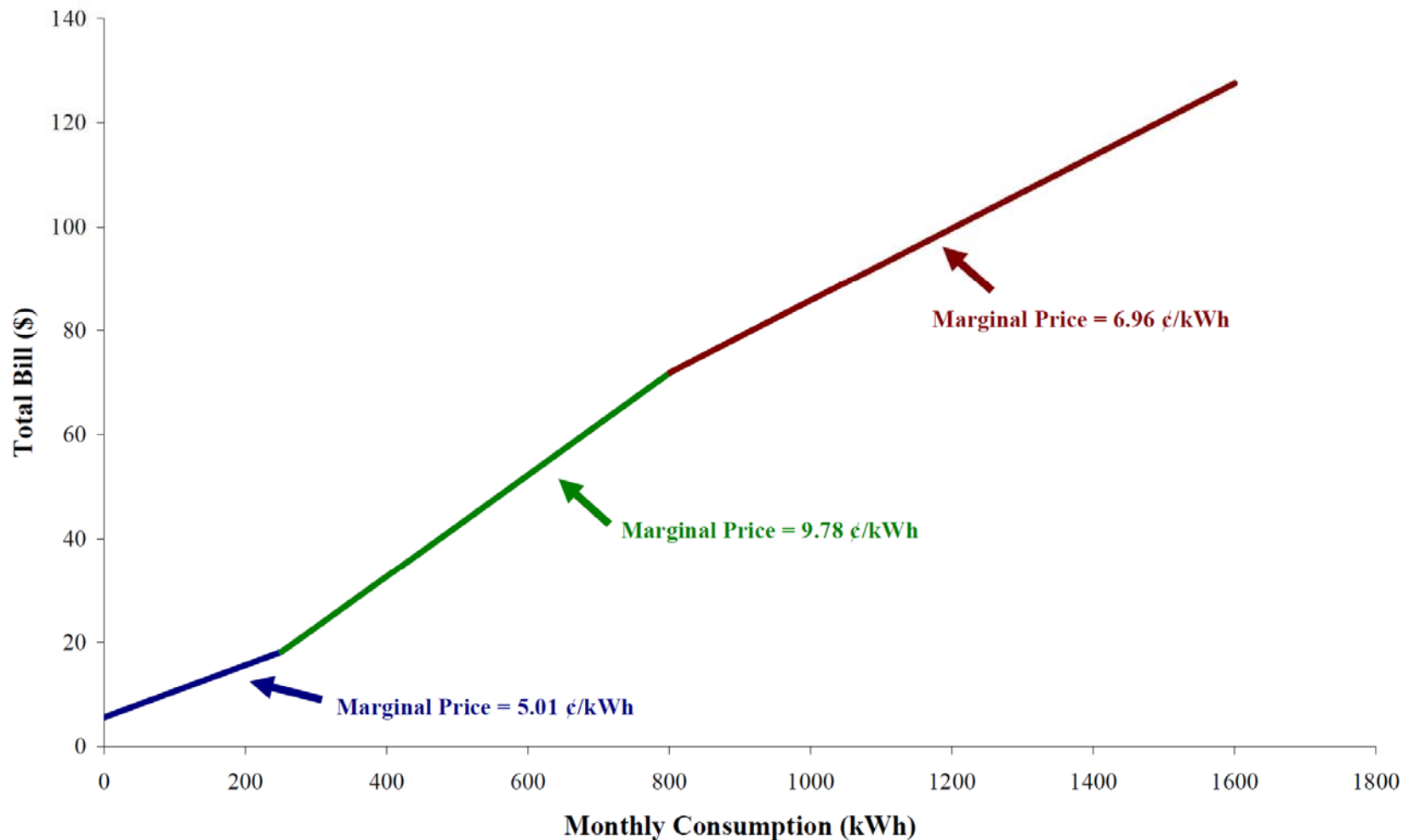
*** FACTURA PAGADA ***

SIN EL SELLO DE CANCELADO, NO TIENE VALOR

ORIGINAL - USUARIO

ORIGINAL - USUARIO

Sample Residential Tariff Function



Front



1 Account Number 1212-1212-12-4 10 21

For less detailed billing information on your monthly bill, check box on right.

5 Current Customer
12345 Your Street
Your City ST 12345-1234

400 00000240764 12121212124 mmddyyy2 00000244377

Page 1 of 2		
Name/Service Address	For Inquiries Call	Account Number
Current Customer 12345 Your Street Your City ST 12345-1234	Duke Energy 1-800-123-4567	1212-1212-12-4

9 Mail Payments To	Account Information	
PO Box 9001076 Louisville KY 40290-1086	Payments after mmm dd not included Last payment received mmm dd	Bill Prepared on mmm dd, yyyy Next meter reading mmm dd, yyyy

Urgent Messages are printed in this section of the bill, with a box around the bill text.

Meter	Number	11 Reading Date	From	To	Days	Previous	Present	Usage
Gas	111111111	mmm dd	mmm dd	mmm dd	31	6266	6397	131
Elec	222222222	mmm dd	mmm dd	mmm dd	31	60377	61451	1,074

13 Gas - Residential
Usage - 131 CCF
Duke Energy - Rate RS \$82.83
Current Gas Charges \$92.83

Current Billing
Balance - Previous Bill \$140.00
Payment(s) Received 140.00 cr
Balance Forward 0.00
Current Gas Charges 92.83
Current Electric Charges 82.31
Current Amount Due \$175.14

15 Electric - Residential
Usage - 1,074 kWh
Duke Energy - Rate RS \$26.29
Quality Electric - Rate 7D 56.02
Current Electric Charges \$82.31

16 PRICE TO COMPARE: In order for an average residential customer to save money, an electric supplier must offer a price lower than x.xx cents per kWh. Your Price to Compare may be different based on your usage. Visit www.duke-energy.com to calculate your individual Price to Compare or contact Duke Energy for a written explanation.

Due Date	Amount Due	After Mar 23, 2003 Pay
Mar 23, 2003	\$175.14	\$177.77



visit us at www.duke-energy.com

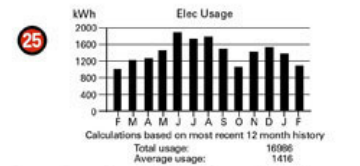
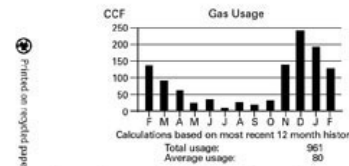
Back



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Name	Service Address	Account Number
Current Customer	12345 Your St Your City ST	1212-1212-12-4

Explanation of Current Charges		
Gas	Duke Energy	
Meter # 111111111	Rate RS - Residential Service	
CCF Usage - 103	Fixed Delivery Service Charge	\$ x.xx
	Usage-Based Charge	
	103 CCF @ \$ x.xxxxxxx	x.xx
Mmm dd - Mmm dd	Rider MSR-G	x.xx cr
31 Days	Gas Delivery Riders	x.xx
	Gas Cost Recovery	
	103 CCF @ \$ x.xxxxxxx	x.xx
		\$ xx.xx
Total Current Gas Charges		\$ xx.xx
Electric	Duke Energy	
Meter # 222222222	Rate RS - Residential Srvc	
kWh Usage - 859	Distribution - Customer Chg	\$ x.xx
	Delivery Charges	
	Distribution - Energy Chg	
	859 kWh @ \$0.01994900	x.xx
Mmm dd - Mmm dd	Rider TCR	x.xx
31 Days	Delivery Riders	x.xx
	Total Delivery Charges	\$ x.xx
	Generation Charges	
	Generation - Energy Chg	
	859 kWh @ \$0.04418000	x.xx
	Rider FPP	x.xx
	Rider AAC	x.xx
	Generation Riders	x.xx
	Total Generation Charges	\$ x.xx
Total Current Electric Charges		\$ xx.xx



	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB
Gas	143	87	58	24	38	7	27	17	29	142	238	191	103
Electric	1025	1209	1211	1478	1855	1689	1698	1528	1035	1376	1575	1352	980



visit us at www.duke-energy.com

Zoom of Back



Page 2 of 2

Name	Service Address	Account Number
Current Customer	12345 Your St Your City ST	1212-1212-12-4

18

Explanation of Current Charges		
Gas Meter # 11111111 CCF Usage - 103 Mmm dd - Mmm dd 31 Days	Duke Energy	
	Rate RS - Residential Service	
	Fixed Delivery Service Charge	\$ x.xx
	Usage-Based Charge	
	103 CCF @ \$ x.xxxxxxx	x.xx
	Rider MSR-G	x.xx cr
	Gas Delivery Riders	x.xx
	Gas Cost Recovery	
	103 CCF @ \$ x.xxxxxxx	x.xx
		\$ xx.xx
Total Current Gas Charges		\$ xx.xx
Electric Meter # 22222222 kWh Usage - 859 Mmm dd - Mmm dd 31 Days	Duke Energy	
	Rate RS - Residential Srvc	
	Distribution - Customer Chg	\$ x.xx
	Delivery Charges	
	Distribution - Energy Chg	
	859 kWh @ \$0.01994900	x.xx
	Rider TCR	x.xx
	Delivery Riders	x.xx
	Total Delivery Charges	\$ x.xx
	Generation Charges	
	Generation - Energy Chg	
	859 kWh @ \$0.04418000	x.xx
	Rider FPP	x.xx
	Rider AAC	x.xx
	Generation Riders	x.xx
	Total Generation Charges	\$ x.xx
Total Current Electric Charges		\$ xx.xx

23

24

19

21

20

22



Electricity summary

Electricity account

	Last reading	This reading	Electricity units used	Cost split	Charges
Tariff - Go Save Electricity / Monthly Direct Debit				Meter: S04C28072	
24 Hour	13/10/11 27350 Customer reading	13/12/11 28472 Actual	1122 kWh	first 157 at 16.650p next 1312 at 10.550p	£26.14 £138.42
24 Hour	13/12/11 28472 Actual	31/12/11 28819 Estimate	347 kWh		
24 Hour	31/12/11 PC - 28819 Estimate	28/03/12 30372 Actual	1553 kWh	first 158 at 17.710p next 1395 at 14.010p	£27.98 £195.44
Your Supply Number				Cost of electricity used this period	£387.98
S 01 801 100 17 1494 4591 301				Monthly Direct Debit Discount	- £40.00
				Subtotal (excluding VAT)	+ £347.98
				VAT at 5.0% on £347.98	+ £17.39
				Total electricity charges this period	£365.37

Your meter reading

* **Price Change**
Please note - there's been a price change during this bill period. We've split your fuel usage over the different prices.

How your Direct Debit account adds up

Your Direct Debit account	Electricity account
Balance on last bill	£183.69 debit
Payment received with thanks on 24/10/11	£35.00 credit
Payment received with thanks on 23/11/11	£98.00 credit
Payment received with thanks on 23/12/11	£98.00 credit
Payment received with thanks on 23/01/12	£98.00 credit
Payment received with thanks on 23/02/12	£98.00 credit
Payment received with thanks on 23/03/12	£98.00 credit
Account balance before charges	£341.31 credit

This bill Electricity account



Reliant Energy Account: [REDACTED]

Page 1 of 3

Account Information

Account #:

Invoice #:

Customer Name:

Billing Date: Jul 17, 2006

Date Due	Amount Due	Amount Due After Due Date
08/02/2006	\$ 422.77	\$ 443.91

Account Summary

Previous Amount Due	\$366.72
Payment 07/03/2006	-366.72
Balance Forward	0.00
Total Current Charges	422.77
Total Due	\$422.77

Electricity Usage Summary

For more usage and temperature information logon to reliant.com/myaccount



Billing Period	06/09/2006 - 07/11/2006	05/10/2006 - 06/09/2006	06/09/2005 - 07/11/2005
Billing Days	32	30	32
Electricity Used (kWh)	2489	2164	3040
Avg. High Temperature*	90	87	94
Avg. Daily Usage (kWh)	78	72	95

*Temperature Source: National Weather Service Region: Coastal Texas

Questions or Comments

Customer Service

www.reliant.com/myaccount
Email us at: service@reliant.com

713-207-7777 24-hours a day
1-866-222-7100 24-hours a day
TDD Device for Hearing Impaired
1-888-467-3542

Reliant Energy Residential Services
Certificate: 10007

Service Address

For outages or emergencies
call 713-207-2222
or toll-free 1-800-332-7143.

ESI ID:

Electric Usage Detail

Meter Number

Current Read 07/11/2006 33541
Previous Read 06/09/2006 31052
kWh Multiplier 1
kWh Usage 2,489

Current Electricity Charges Detail

32 Day Billing Period From 06/09/2006 To 07/11/2006

Residential Service

Reliant Energy Electric Service		
Monthly Customer Charge		5.59
First 250 kWh	250 kWh @ \$0.029441/kWh	7.36
All Additional kWh	2,239 kWh @ \$0.077171/kWh	172.79
Fuel Factor for Generation	2,489 kWh @ \$0.092718/kWh	230.78
City Sales Tax 1.50%		6.25
Total Current Charges		\$422.77
The average price you paid for electric service this month (per kWh)		\$0.167



Florida Power & Light Company
PO Box 025576
Miami, FL 33102

/ 00

00000000000000000000000000000000

Please request changes on the back.
Notes on the front will not be detected.

The amount enclosed includes the following donation:

FPL Care To Share \$ _____

B 8 8888 8

AUTO **CO 0000
0 000000

JANE CUSTOMER
123 ANY ST
ANYTOWN FL 33000-0000

Make check payable to FPL in U.S. funds
and mail along with this coupon to:

FPL
GENERAL MAIL FACILITY
MIAMI FL 33188-0001



Account number	Total amount you owe	New charges due by	Amount enclosed
12345-67890	\$147.36	Nov 08 2010	\$

Your electric statement

For: Sep 17 2010 to Oct 18 2010 (31 days)

Customer name: JANE CUSTOMER

Service address: 123 ANY ST

Account number: 12345-67890

Statement date: Oct 18 2010

Next meter reading: Nov 16 2010

Amount of your last bill	Payments (-)	Additional activity (+ or -)	Balance before new charges (=)	New charges (=)	Total amount you owe (=)	New charges due by
156.73	156.73 CR	0.00	0.00	147.36	\$147.36	Nov 08 2010

Meter reading - Meter 5CXXXXX

Current reading 79065
Previous reading - 77725
kWh used 1340

Amount of your last bill 156.73
Payment received - Thank you 156.73CR
Balance before new charges \$0.00

Energy usage

	Last Year	This Year
kWh this month	1620	1340
Service days	29	31
kWh per day	56	43

New charges (Rate: RS-1 RESIDENTIAL SERVICE)

Electric service amount	127.35**
Storm charge	1.57
Gross receipts tax	3.31
Franchise Charge	5.95
Utility Tax	9.18
Total new charges	\$147.36

**The electric service amount includes the following charges:

Customer charge: \$5.90
Fuel: \$55.08

(First 1000 kWh at \$0.038570)

(Over 1000 kWh at \$0.048570)

Non-fuel: \$66.37

(First 1000 kWh at \$0.046990)

(Over 1000 kWh at \$0.056990)

Total amount you owe \$147.36

- Payment received after **November 08, 2010** is considered **LATE**; a late payment charge of **1.50%** will apply and your account may be subject to an adjusted deposit billing.

Bill Saliency

- Some bills don't even have sufficient information to determine marginal price
- Bills tend to display total expenditures and “breakdown of expenditures” more saliently than marginal price schedule

Saliency of Retail Energy

Marginal Prices vs. Expenditures

Electricity & Gas

 Florida Power & Light Company PO Box 025576 Miami, FL 33102	/	00	00000000000000000000000000000000
Please request changes on the back. Notes on the front will not be detected.			The amount enclosed includes the following donation: FPL Care To Share \$ _____
B 8	8888 8		
AUTO **CO 0000 0 000000			
JANE CUSTOMER 123 ANY ST ANYTOWN FL 33000-0000			Make check payable to FPL in U.S. funds and mail along with this coupon to:
			FPL GENERAL MAIL FACILITY MIAMI FL 33188-0001

Your electric statement

For: Sep 17 2010 to Oct 18 2010 (31 days)
Customer name: JANE CUSTOMER
Service address: 123 ANY ST

Account number: 12345-67890

Statement date: Oct 18 2010
Next meter reading: Nov 16 2010

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Meter reading - Meter 5CXXXXX

Current reading	79065
Previous reading	- 77725
kWh used	1340

Amount of your last bill	156.73
Payment received - Thank you	156.73CR
Balance before new charges	\$0.00

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Gasoline



A Field Experiment Underway Now



INFORMACION IMPORTANTE

Ahorre Energía y Ahorre Dinero

Estimado Cliente:

La tarifa eléctrica establecida por el CONELEC funciona de manera escalonada. Esto implica que hay un incremento importante en su factura mensual si usted consume más de 110 kWh.

La siguiente información con respecto a su consumo mensual de electricidad durante el pasado año puede ser de su interés.

Su consumo mensual fue:

115 kWh

Un hogar similar al suyo consume en promedio:

110 kWh

Durante el pasado año usted pagó cada mes alrededor de US\$ 12.56 por su consumo de electricidad. Si usted reduce su consumo mensual de electricidad en 5 kWh (alrededor de un 4% de su consumo actual), usted reduciría su factura mensual de electricidad cerca de un **43%** y **ahorraría aproximadamente US\$ 64 por año**. Le exhortamos que haga un uso eficiente de la energía para ahorrar dinero.

Lea con atención los consejos para ahorrar energía que le damos a continuación y empiece a ahorrar dinero ya! Comparta esta información con todos los demás miembros del hogar.

¡AHORRE ELECTRICIDAD, AHORRE DINERO!

Standard Arsenal of Policy Instruments

1. Price
2. Behavioral Interventions / Nudges
 - ❑ Energy/water conservation
 - ❑ Smoking cessation
 - ❑ Voting
 - ❑ Charitable donations
 - ❑ More...

The Short-Run and Long-Run Effects of Behavioral Interventions: Experimental Evidence from Energy Conservation

Forthcoming *American Economic Review*

By Allcott and Rogers

(some slides borrowed from Hunt Allcott)

The Information Intervention: Home Energy Reports



Home Energy Report

Account number: 1234567890
Report period: 11/09/09 – 12/09/09

We are pleased to provide this personalized report to help you save energy.

The purpose of the report is to:

- Provide information
- Help you track your progress
- Share energy efficiency tips



This information and more is available at UtilityCo.com/reports

BOB SMITH
414 NICOLLET MALL GO 6
MINNEAPOLIS, MN 55401

Last Month Neighbor Comparison | You used **19% MORE** energy than your efficient neighbors.



* This energy index combines electricity (kWh) and natural gas (therms) into a single measurement.

How you're doing:

Great 😊😊

► **GOOD** 😊

More than average

Who are your Neighbors?

All Neighbors

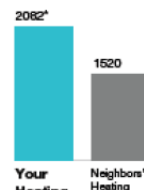
Approximately 100 occupied, nearby homes that are similar in size to yours (avg 2,109 sq ft) and have both electricity and natural gas service

Efficient Neighbors

The most efficient 20 percent from the "All Neighbors" group

Last Winter Heating Comparison

! Last winter, you used **37% more energy on heating** than your neighbors.



Best ways to save this winter:



Lower your thermostat before leaving home

Adjusting by 10° can save 10% on heating costs.



Seal air leaks around windows and doors

Use caulk or weatherstripping to eliminate cold drafts.



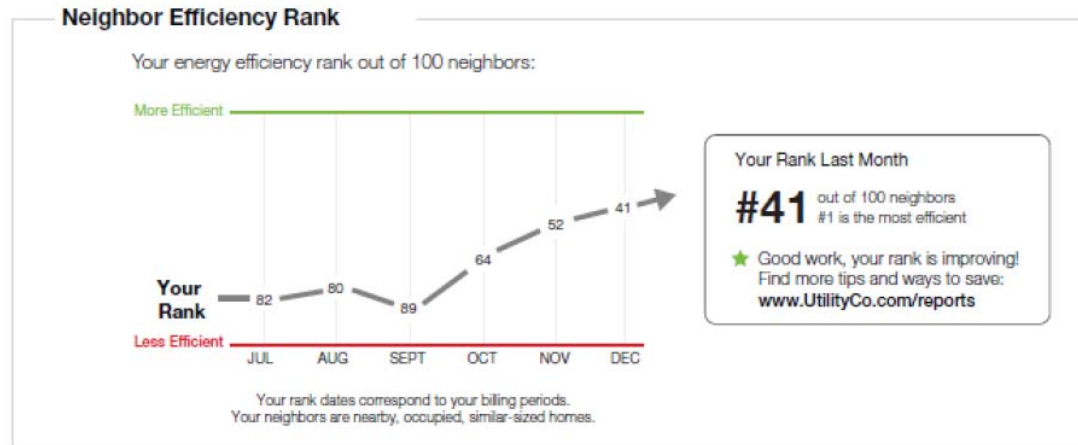
Let sunshine in for warmth

Open blinds during the day to capture free heat.

For more tips, visit UtilityCo.com/reports



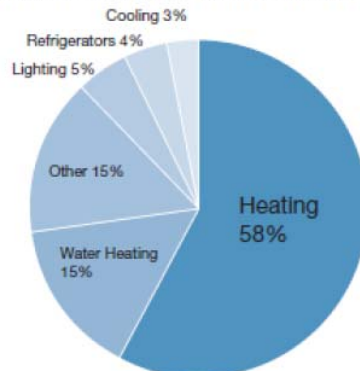
Home Energy Reports (Page 2)



Understanding Energy Use

Heating is the largest use of energy for a typical household in the East Metro area, accounting for more than 50% of total energy use. To maximize your savings, focus on the biggest users first.

Typical annual energy use in the East Metro area



Other appliances and electronics include dishwashers, washing machines, dryers, computers, TVs and entertainment systems.
 Based on a typical household with gas heating & water heating.

Top Tips For Saving

	Save up to
☐ Look for the ENERGY STAR® label Next Steps: Look for the ENERGY STAR label when shopping for appliances and electronics.	\$600/yr
☐ Improve insulation and seal air leaks Next Steps: Start with the places easiest to access, such as an attic.	\$305/yr
☐ Seal leaky ducts Next Steps: Use mastic (a special adhesive) or duct tape to seal all accessible duct joints.	\$170/yr
☐ Recycle your second refrigerator Next Steps: Try rearranging your main fridge to fit everything from your second fridge.	\$145/yr
☐ Turn off computer at night Next Steps: Program your computer to automatically turn off after periods of inactivity.	\$75/yr
☐ Set your thermostat wisely Next Steps: Set your thermostat 10 degrees off from your preferred setting when you're away or sleeping.	\$65/yr
☐ Install efficient showerheads Next Steps: Get a new efficient showerhead and bathroom faucet aerator for free! Visit xcelenergy.com/energyreport for details.	\$45/yr

Already do these tips?
 Find more ways to save online

For energy-saving tips visit
www.UtilityCo.com/reports

Primary Research Questions

- Note this is “only” an information treatment. No changes in prices.
- Possible mechanisms:
 - Information → re-optimized behavior (permanent)
 - Information → cue to change behavior (possible backsliding)
- 1. Do the HER behavioral interventions reduce electricity consumption?
 - Yes, within days.
 - But “Action and backsliding” in the short-run
- 1. Is the reduction persistent?
 - When reports discontinue, long-run effects decay slowly.

Key Features of Research Design

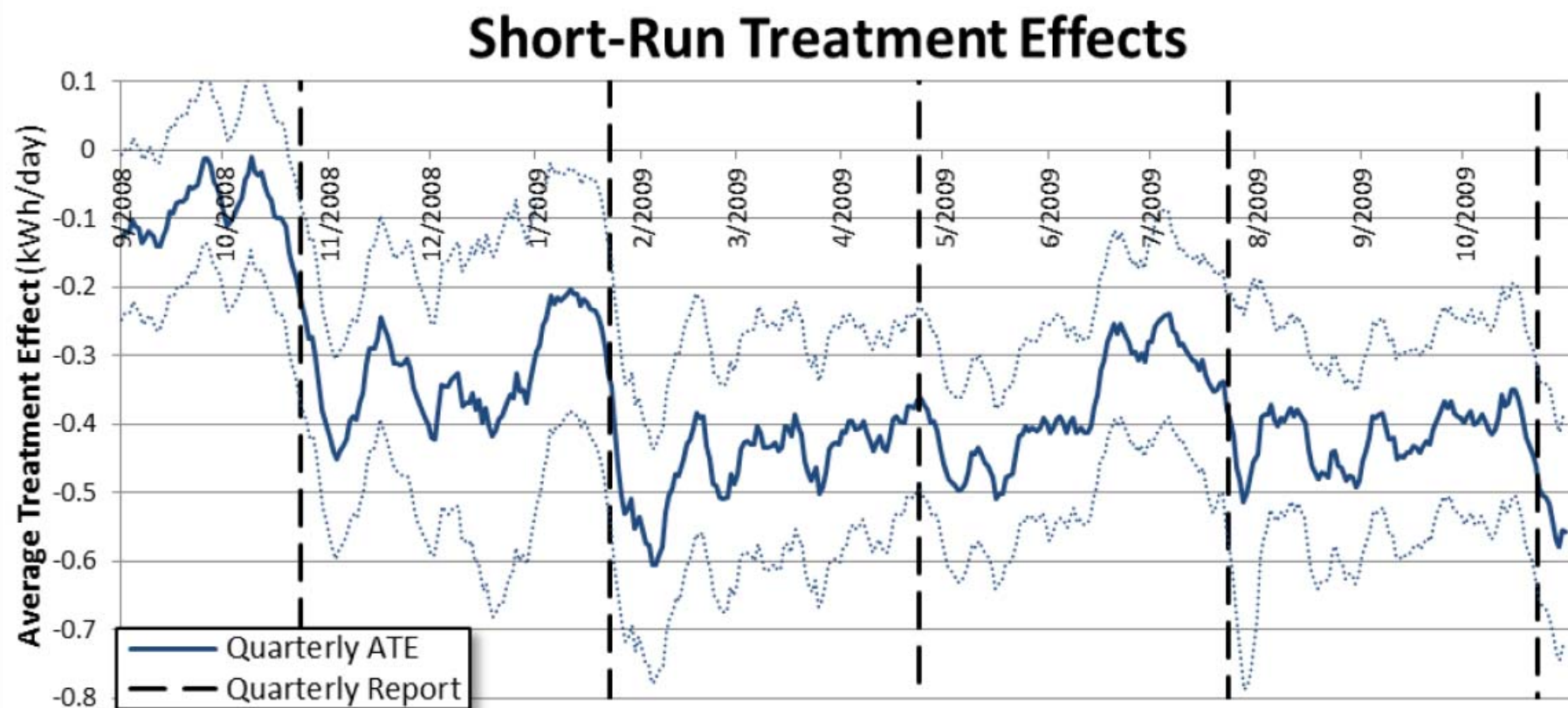
- 4 years of daily electricity consumption data on 122,000 households following repeated random interventions
- Treatment begins October 2008
 - 79,000 households
 - Three groups: monthly, quarterly, control
- September 2010
 - 11,600 households randomly dropped from treatment

Summary Stats and Balance

Baseline Usage (kWh/day)

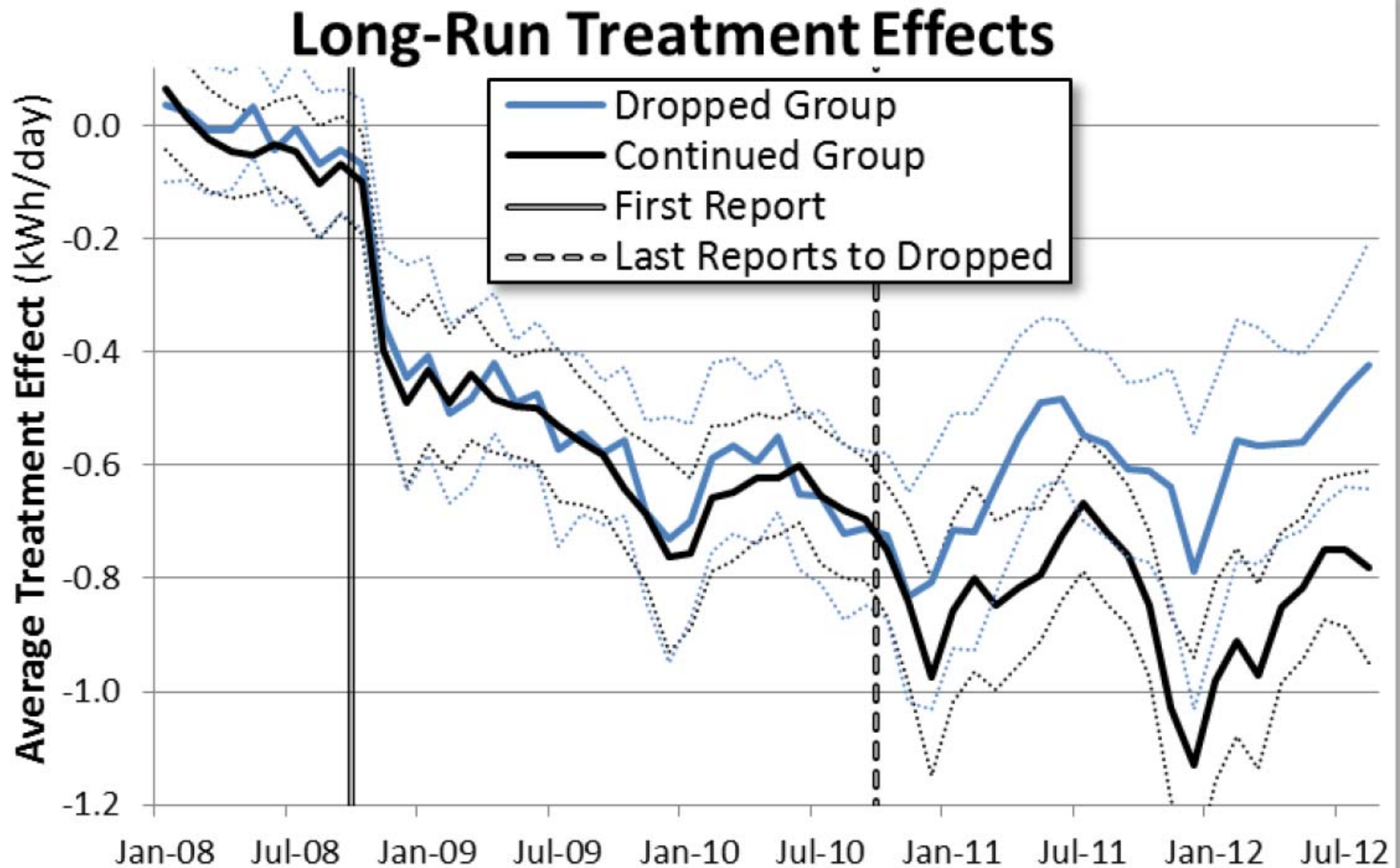
Mean	30.3
Standard deviation	13.50
Treatment - Control	0.045
(Standard error)	(0.097)
Dropped - Continued	0.062
(Standard error)	(0.154)

Graphical Snapshots of Findings



Level of observation: Daily Data

Graphical Snapshots of Findings



Future Directions Related to Allcott & Rogers?

1. What is the “active ingredient”?
 - Social comparison? Energy conservation tips? Personalized energy use feedback?
2. What is the mechanism?